

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Purchases

18

Features

37

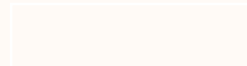
Missing Values

Review Rating column

\$59.76

Avg Purchase

Demographics, purchase details, shopping behavior — loaded into PostgreSQL for analysis.



Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean

Imputed Review Rating by category median

03

Engineer

age_group, purchase_frequency_days

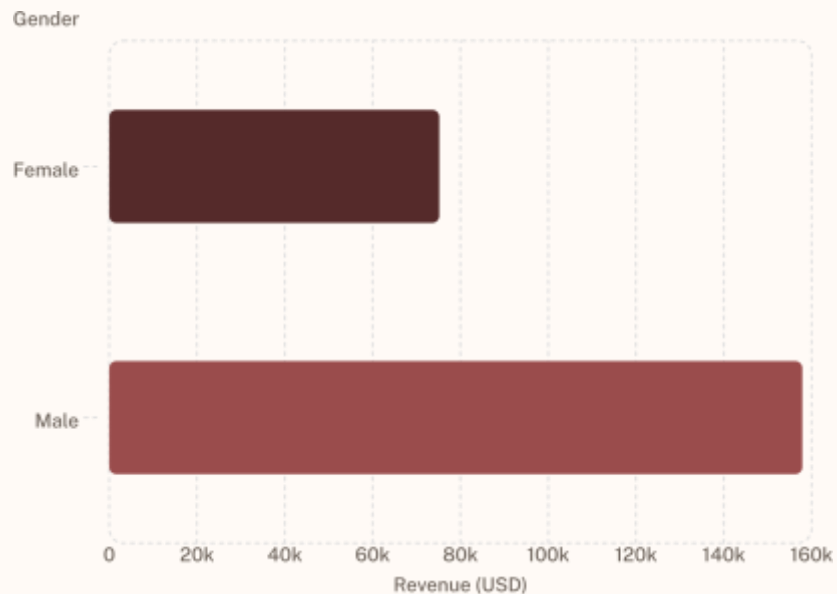
04

Load to DB

Connected to MySQL



Revenue by Gender

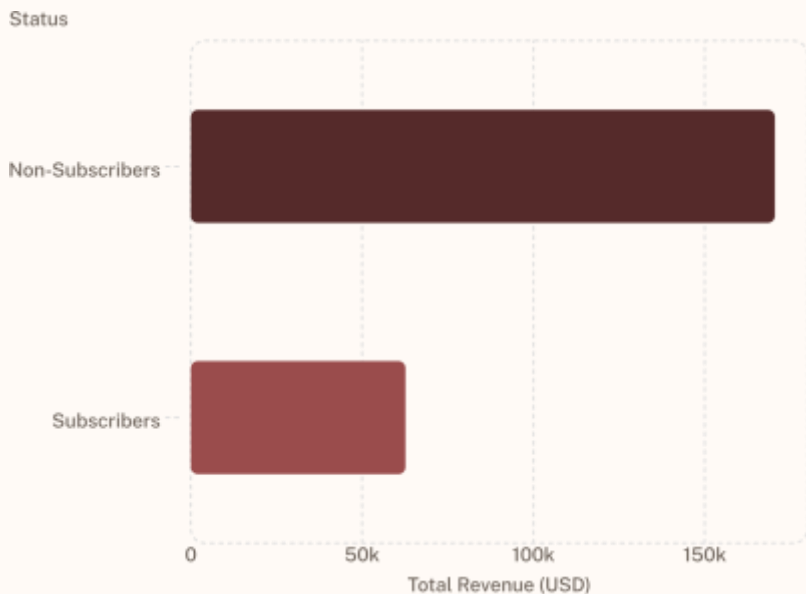


Male customers drive 2× revenue

Male: **\$157,890** vs. Female: **\$75,191**

68% of customers are male — a key demographic to target.

Subscribers vs. Non-Subscribers



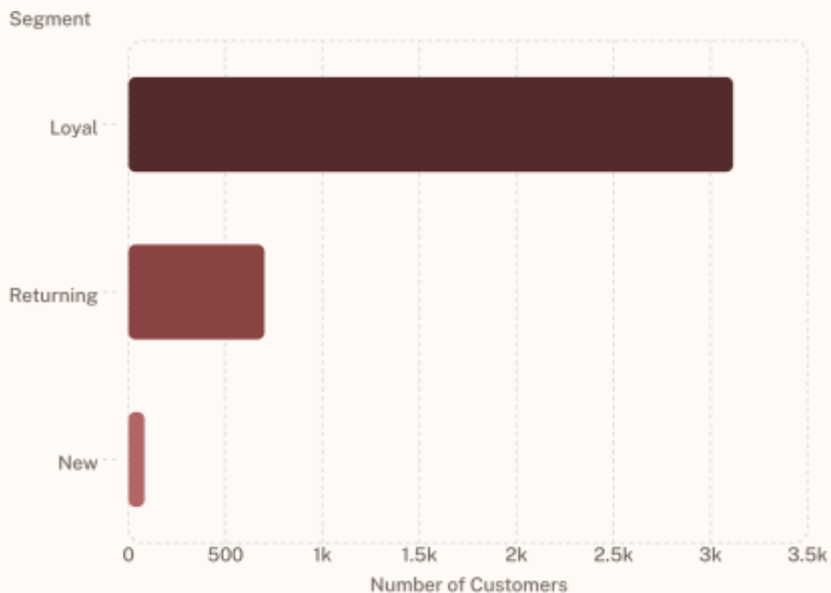
73% are non-subscribers

Non-subscribers generate **\$170K** vs. subscribers **\$62K**

Only 27% subscribed — significant growth opportunity.

Repeat buyers: 2,518 non-subscribers vs. 958 subscribers

Customer Segmentation

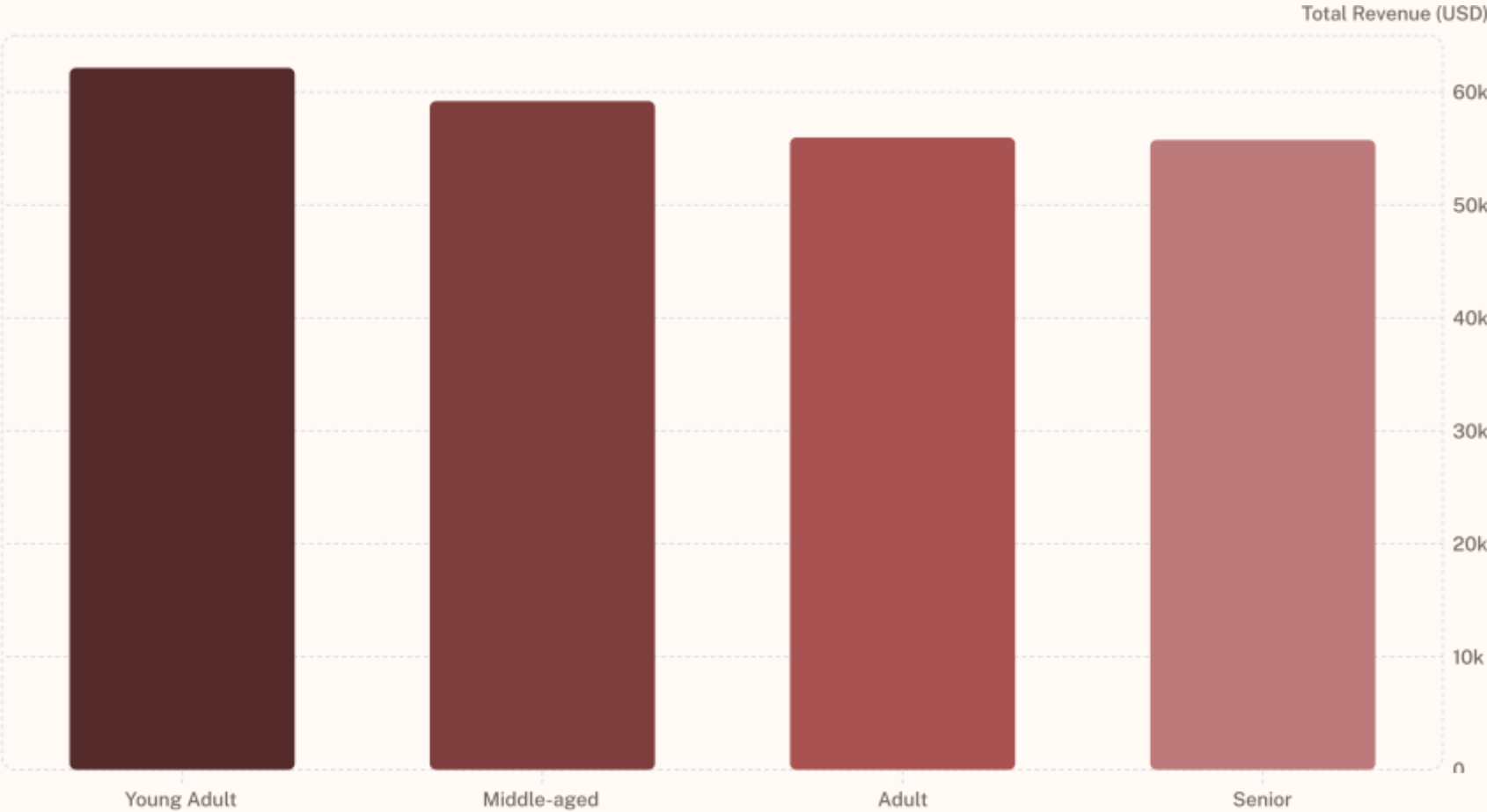


Loyal customers dominate

80% classified as Loyal — strong retention base.

Only 83 New customers — acquisition needs focus.

Revenue by Age Group



Top Products & Discount Patterns

Top 5 by Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Product	Discount %
Hat	50%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Interactive filters: Gender, Category, Shipping Type, Age Group —
Revenue & Sales views.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales lift vs. margins



Targeted Marketing

Focus on Young Adults & express-shipping users

